

Mad 1 Project Report

1. Student Details

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About Me: I am a student in the IIT Madras BS Degree program with a strong interest in technology and web development. I enjoy learning new tools and building applications that solve real world problems.

2. Project Details

Project Title: Placement Portal Application

Problem Statement:

To design and build a web based placement management system that connects students with job opportunities, allows companies to post and manage hiring drives, and enables administrators to monitor and verify the placement process.

Approach:

The application was built using the Flask framework with a role based system for **Admin, Student, and Company** users. Admin verifies companies and placement drives before they become available to students. Students can browse verified drives, upload resumes, apply for jobs, and track their application status. Companies can create placement drives and review student applications. The system uses SQLAlchemy ORM for database management and supports searching and tracking of applications within the portal.

3. AI/LLM Declaration

I used **ChatGPT (GPT 5.3)** to assist with Bootstrap styling classes , code suggestions, and guidance for implementing resume file upload and download using SQLAlchemy.

The extent of AI/LLM usage is approximately **12–18%**, mainly for code suggestions and documentation support.

All core implementation, including **database schema design, route development, application logic, and verification workflows**, was implemented manually.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	For backend routing and the request handling
SQLAlchemy	Object Relational Mapper for SQLite database
SQLite	Lightweight local database for storing user data
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap 5	For buttons , form and table templates
Sqlalchemy.or_	OR condition for searching
HTML/CSS	User interface and user based styling
Python Datetime	Managing drive deadline and application date

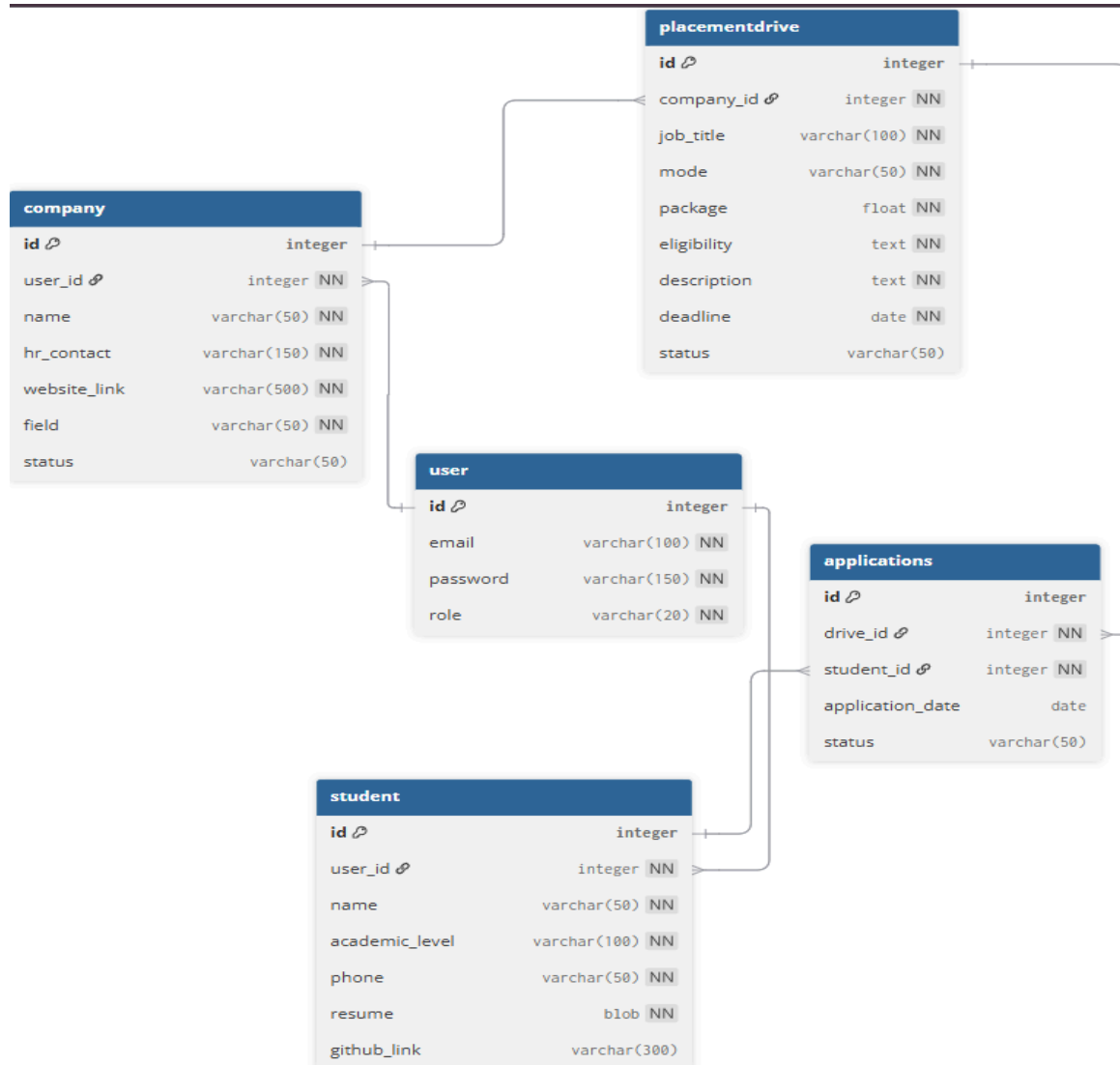
5. Database Schema / ER Diagram

Tables:

1. **User** — stores user authentication credential and role (id, email, password,role)
2. **Company** —store company information (id, user_id, name, hr_contact, website_link, field, status)
3. **Student** —stores student information (id, user_id, name, academic_level, phone, resume, github_link)
4. **PlacementDrive** — Stores job posting details (id, company_id, job_title, mode, package, eligibility, description, deadline, status)
5. **Applications** —Tracks student applications to placement drives (id, drive_id, student_id, application_date, status)

Relationships:

- One-to-One → User → Student
- One-to-One → User → Company
- One-to-Many → Company → PlacementDrive
- One-to-Many → PlacementDrive → Applications
- One-to-Many → Student → Applications



6. API Resource Endpoints/Routings

Route	Methods	Description
/	GET	Home page and role selection
/student-register	GET, POST	Student registration with resume upload
/student-dashboard/<student_id>	GET	Student dashboard with available drives and application status
/student-apply-placement-drive/<student_id>/<drive_id>	GET	Submit application to placement drive
/admin-dashboard	GET	Admin dashboard with statistics and verification workflow
/admin-approve-placement-drive/<drive_id>	GET	Approve placement drive (verification system)
/add-drive/<company_id>	GET, POST	Company creates new placement drive
/drive-applications/<drive_id>	GET	Company views applications for specific drive

7. Architecture and Features (optional)

Architecture Overview:

- **app.py** – Main Flask application and initialization
- **application/database.py** – Database configuration using SQLAlchemy
- **application/model.py** – Database models (User, Student, Company, PlacementDrive, Applications)
- **application/controllers.py** – Route handlers
- **templates/** – Jinja2 HTML templates organized by user role
- **static/css** – CSS files
- **static/images** – images
- **instance/** – SQLite database file

Implemented Features:

Role-based authentication for Admin, Student, and Company

Admin verification system for companies and placement drives

Student features - register, upload resume, search drives, apply for jobs, track applications

Company features – create placement drives and manage applicants

Profile management for student details and resume updates

Application status tracking (Applied, Shortlisted, Selected, Rejected)

Dashboard views for each user role with relevant information.

8. Video Presentation

Drive Link:

https://drive.google.com/file/d/12foT9fFracjJS7SZMkEA_W9mWiwj1q3n/view?usp=sharing